writeup

Anonymous ACL submission

Abstract

(Abstract goes here.)

1 Experiment 1: UP Independently

1.1 Items

Placeholder reference: Houghton et al. (2020).

The classifier was trained to distinguish the hidden-state representation of the particle *up* from representations of other tokens in the same sentence. Positive training examples consisted of 1,000 occurrences of standalone *up* (i.e., functioning as a verbal particle, as in *pick up*), drawn from sentences in the C4 corpus. Negative examples were 1,000 tokens randomly selected from the same sentences, restricted to tokens whose decoded string consists entirely of alphabetic characters (no numbers, punctuation, or special characters), and excluding *up* itself and any token containing *up* as a substring. The validation set was constructed identically from a non-overlapping set of sentences (targeting 1,000 positive, 1,000 negative; exact counts vary slightly by model depending on how many sentences yield a valid token position under each tokenizer — see Table 2).

The classifiers targeting OLMo-3 7B and the BabyLM models were trained on the **same underlying sentences**, but the frequency and predictability statistics used in the analyses were derived from the respective corpus for each model (the C4 corpus for the OLMo model and the BabyLM corpus for the BabyLM models). In other words, the training sentences were identical across models, but the tokenization and corpus statistics varied.

Classifier training and validation split sizes are shown in Table 2 (Appendix). The test set comprised verb–*up* phrasal verb types (e.g., *pick up*, *set up*) with at least 20 occurrences in the corpus; up to 20 sentences were sampled per type. Corpus frequency and forward transitional probability (FTP; $P(\text{up} \mid V)$) were derived from the corpus matched to each model's training distribution. Because the BabyLM corpus is smaller than C4, fewer V+up types attain a valid (non-zero) FTP estimate under BabyLM statistics; only types with a valid FTP are included in the analyses. Full item-level statistics are reported in Table 1 (Appendix).

For OLMo-3 7B, 3,927 unique V+*up* types were included, with a median corpus frequency of 92 (range: 20–390,086) and a median logit(FTP) of -5.38 (range: -10.42–3.87). The BabyLM models shared 1,039 items; these items had a higher median frequency (920) and higher median logit(FTP) (-3.40) under BabyLM-corpus statistics, reflecting the smaller and more concentrated vocabulary of the BabyLM training data.

1.2 Joint model — predicted surfaces (linear)

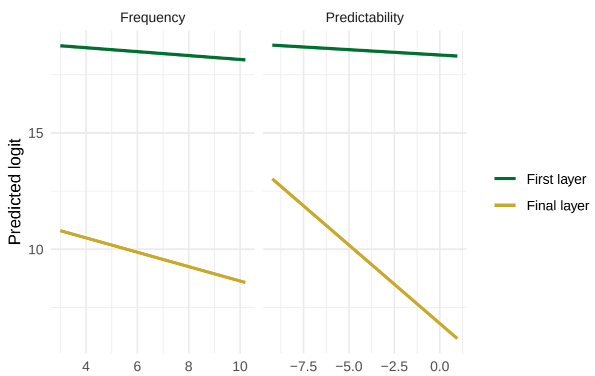

Figure 1: Predicted logit as a function of frequency (left) and predictability (right) from the linear joint brms model, at the first and final layers, for OLMo-3 7B (UP independently). Each line slices the joint surface at the mean of the other predictor.

2 Experiment 2: UP as Subword

2.1 Items

The training set for this experiment combined two types of positive example: 1,000 occurrences of standalone *up* (identical to Experiment 1) and 1,000

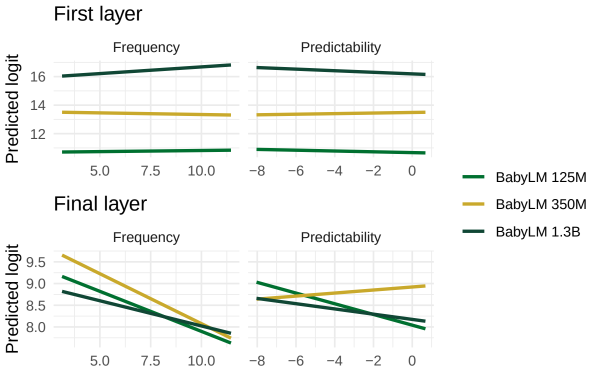

Figure 2: Predicted logit as a function of frequency (left) and predictability (right) from the linear joint brms model at the first (top) and final (bottom) layers, for BabyLM models (UP independently). Each line slices the joint surface at the mean of the other predictor.

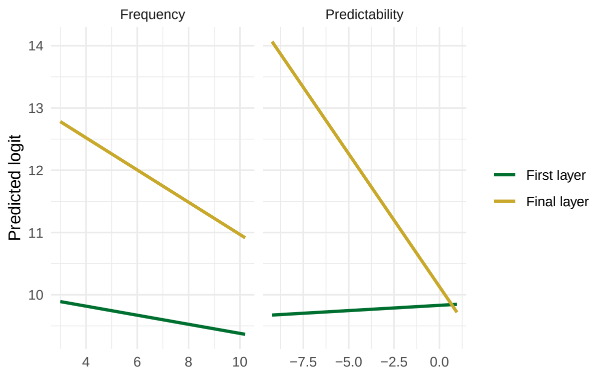

Figure 3: Predicted logit as a function of frequency (left) and predictability (right) from the linear joint brms model, at the first and final layers, for OLMo-3 7B (UP as subword). Each line slices the joint surface at the mean of the other predictor.

occurrences of *up* embedded within a larger word (e.g., *update*, *upon*, *setup*). Critically, the subword positives were restricted to **unique word types**: each up-containing word form contributed exactly one instance, so the classifier could not learn to recognise a high-frequency form like *setup* from repeated exposure. Negative examples (1,000 drawn from each sentence pool) were tokens consisting entirely of alphabetic characters that did not contain *up* as a substring. The validation set was constructed by the same procedure (targeting 1,000 positive, 1,000 negative per sentence pool; exact counts vary by tokenizer — see Table 5). As in Experiment 1, OLMo-3 7B and BabyLM models used the same underlying sentences with their own tokenizers, and BabyLM models used BabyLM-corpus statistics for frequency and FTP. The test set was identical to Experiment 1: all verb–*up* phrasal verb types with at least 20 corpus occurrences, sampled at up to 20 sentences per type.

Classifier training and validation split sizes are shown in Table 5 (Appendix). The OLMo training set contained 1,000 unique up-within-word types; BabyLM models similarly contributed 1,000 unique types, with the exact set varying by tokenizer. Validation-set sizes differ slightly across models because some sentences could not be resolved to a valid token position under each model's tokenizer. The test set is identical to Experiment 1; item-level statistics are shown in Table 1.

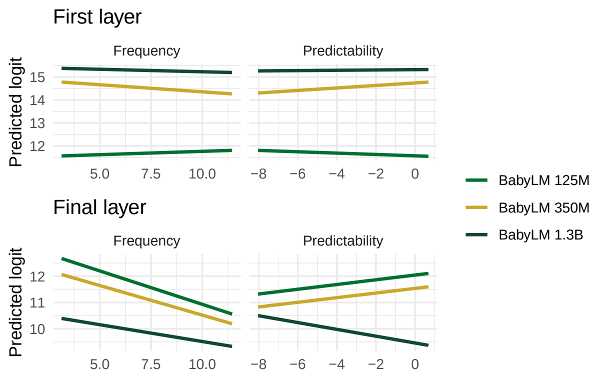

Figure 4: Predicted logit as a function of frequency (left) and predictability (right) from the linear joint brms model at the first (top) and final (bottom) layers, for BabyLM models (UP as subword). Each line slices the joint surface at the mean of the other predictor.

2.2 Joint model — predicted surfaces (linear)

3 Experiment 3: Whisper (ASR Model)

3.1 Items

The Whisper experiment used spoken language data drawn from a mixed-source audio metadata file (`up-audio-metadata.csv`) containing 224,118 timestamped speech segments in which *up* occurred. Word-level timestamps within each segment were obtained using WhisperX forced alignment. Each occurrence of *up* was classified by running spaCy part-of-speech tagging on the full segment transcript: if the token immediately preceding *up* was tagged as a VERB, the instance was labelled *V+up* (e.g., *pick up*, *clean up*); otherwise it was labelled *word_up* (e.g., *it up*, *them up*). This classification procedure mirrors the one used to identify standalone *up* in Experiments 1 and 2. Of the 224,118 segments, 58,751 were classified as *word_up* and 165,367 as *V+up*, spanning 4,161 unique V+up types.

The **training and validation sets** used 1,000 *word_up* instances each as positive examples, paired with 1,000 randomly selected non-*up* words from the same segment as negative examples. This mirrors the standalone-*up* training design of Experiments 1 and 2. Negative words were restricted to those whose string consists entirely of alphabetic characters (matching the `token.isalpha()` criterion used in Experiments 1 and 2), with a minimum duration of 20 ms. Training and validation segments were non-overlapping.

The **test set** comprised V+up phrasal verb types with at least 5 occurrences in the audio dataset, yielding 1,466 qualifying types, with up to 20 instances per type (median: 16 sentences per type; range: 5–20). The minimum-occurrence threshold is lower than in Experiments 1 and 2 (which required at least 20 corpus sentences per type), reflecting the sparser coverage of audio relative to written text. Corpus frequency and FTP values were drawn from the C4-derived lookup table used in Experiments 1 and 2. Classifier training and validation split sizes are shown in Table 8 (Appendix) and full item-level statistics are shown in Table 1 (Appendix). After filtering to types with a valid C4 FTP estimate, 1,397 unique V+*up* types were retained for both encoder and decoder, with a median corpus frequency of 683 (range: 20–390,086) and a median logit(FTP) of -4.05 (range: -10.42–3.87).

3.2 Joint model — predicted surfaces (linear)

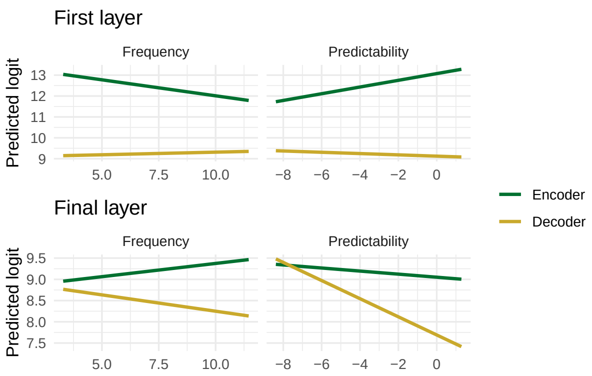

Figure 5: Predicted logit as a function of frequency (left) and predictability (right) from the linear joint brms model at the first (top) and final (bottom) layers, for Whisper encoder and decoder. Each line slices the joint surface at the mean of the other predictor.

References

Zachary N. Houghton, Misaki Kato, Melissa M. Baese-Berk, and Charlotte Vaughn. 2020. Perceptual consequences of pauses: Credibility and fluency of native and non-native speech. *The Journal of the Acoustical Society of America*, 148(4_Supplement):2762–2762.

4 Appendix

4.1 Test Set Items

Model	N items	Med. freq	Freq range	Med. predic.	Predic. range
OLMo-3 7B	3927	92	20–390,086	-5.38	-10.41–3.89
BabyLM	1039	920	20–390,086	-3.40	-10.38–1.99
Whisper encoder	1397	683	20–390,086	-4.05	-10.41–3.89
Whisper decoder	1397	683	20–390,086	-4.05	-10.41–3.89

Table 1: Test set statistics for all experiments. Only items with a valid (non-zero) predictability estimate are included. Frequency is the raw corpus count; predictability is logit(FTP). BabyLM models (125M, 350M, 1.3B) share identical items and corpus statistics. Whisper encoder and decoder evaluate the same audio items with C4-derived corpus statistics.

4.2 Experiment 1: UP Independently

Model	Train pos	Train neg	Val pos	Val neg
OLMo-3 7B	1,000	1,000	992	1,000
BabyLM 125M	1,000	1,000	999	1,000
BabyLM 350M	1,000	1,000	999	1,000
BabyLM 1.3B	1,000	1,000	916	1,000

Table 2: Classifier training and validation split sizes for Experiment 1 (UP independently). Positive (pos) = standalone *up*; negative (neg) = other alphabetic token from the same sentence.

Model	Coef	Est.	Err.	95% CI	% > 0
OLMo-3 7B	Intercept	18.58	0.01	[18.56, 18.60]	100.0
OLMo-3 7B	Freq	-0.14	0.01	[-0.16, -0.12]	0.0
OLMo-3 7B	Predic.	-0.11	0.01	[-0.13, -0.09]	0.0
OLMo-3 7B	Freq:Predic.	-0.09	0.01	[-0.11, -0.07]	0.0
BabyLM 125M	Intercept	10.77	0.01	[10.75, 10.79]	100.0
BabyLM 125M	Freq	0.03	0.01	[0.01, 0.05]	99.3
BabyLM 125M	Predic.	-0.05	0.01	[-0.07, -0.03]	0.0
BabyLM 125M	Freq:Predic.	-0.02	0.01	[-0.04, 0.01]	7.0
BabyLM 350M	Intercept	13.41	0.01	[13.40, 13.43]	100.0
BabyLM 350M	Freq	-0.04	0.01	[-0.06, -0.03]	0.0
BabyLM 350M	Predic.	0.04	0.01	[0.02, 0.05]	100.0
BabyLM 350M	Freq:Predic.	-0.01	0.01	[-0.02, 0.01]	18.3
BabyLM 1.3B	Intercept	16.38	0.03	[16.33, 16.44]	100.0
BabyLM 1.3B	Freq	0.18	0.03	[0.13, 0.23]	100.0
BabyLM 1.3B	Predic.	-0.10	0.03	[-0.15, -0.05]	0.0
BabyLM 1.3B	Freq:Predic.	0.04	0.03	[-0.01, 0.10]	94.5

Table 3: Joint frequency × FTP model at the first layer for UP independently. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Coef	Est.	Err.	95% CI	% > 0
OLMo-3 7B	Intercept	10.27	0.04	[10.18, 10.35]	100.0
OLMo-3 7B	Freq	-0.54	0.05	[-0.63, -0.45]	0.0
OLMo-3 7B	Predic.	-1.56	0.04	[-1.64, -1.48]	0.0
OLMo-3 7B	Freq:Predic.	0.00	0.04	[-0.08, 0.08]	50.7
BabyLM 125M	Intercept	8.47	0.04	[8.39, 8.54]	100.0
BabyLM 125M	Freq	-0.35	0.04	[-0.42, -0.27]	0.0
BabyLM 125M	Predic.	-0.22	0.04	[-0.30, -0.15]	0.0
BabyLM 125M	Freq:Predic.	-0.05	0.04	[-0.13, 0.03]	11.3
BabyLM 350M	Intercept	8.80	0.04	[8.73, 8.87]	100.0
BabyLM 350M	Freq	-0.43	0.04	[-0.50, -0.36]	0.0
BabyLM 350M	Predic.	0.06	0.04	[-0.01, 0.14]	95.2
BabyLM 350M	Freq:Predic.	-0.08	0.04	[-0.15, -0.00]	2.2
BabyLM 1.3B	Intercept	8.38	0.04	[8.30, 8.46]	100.0
BabyLM 1.3B	Freq	-0.22	0.04	[-0.30, -0.13]	0.0
BabyLM 1.3B	Predic.	-0.11	0.04	[-0.19, -0.02]	0.7
BabyLM 1.3B	Freq:Predic.	-0.07	0.05	[-0.17, 0.02]	5.5

Table 4: Joint frequency × FTP model at the final layer for UP independently. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

4.3 Experiment 2: UP as Subword

Model	Train standalone	Train subword	Train neg	Val rows
OLMo-3 7B	1,000	1,000	2,000	3,868
BabyLM 125M	1,000	1,000	2,000	3,582
BabyLM 350M	1,000	1,000	2,000	3,582
BabyLM 1.3B	1,000	1,000	2,000	3,773

Table 5: Classifier training and validation split sizes for Experiment 2 (UP as subword). Positive examples: 1,000 standalone *up* + 1,000 unique up-within-word types. Negative examples: 1,000 from each sentence pool. Val sizes vary by model due to tokenizer coverage.

Model	Coef	Est.	Err.	95% CI	% > 0
OLMo-3 7B	Intercept	9.74	0.01	[9.72, 9.75]	100.0
OLMo-3 7B	Freq	-0.13	0.01	[-0.14, -0.11]	0.0
OLMo-3 7B	Predic.	0.04	0.01	[0.02, 0.05]	100.0
OLMo-3 7B	Freq:Predic.	-0.02	0.01	[-0.03, -0.00]	0.7
BabyLM 125M	Intercept	11.67	0.02	[11.63, 11.72]	100.0
BabyLM 125M	Freq	0.05	0.02	[0.01, 0.10]	99.0
BabyLM 125M	Predic.	-0.05	0.02	[-0.10, -0.01]	1.1
BabyLM 125M	Freq:Predic.	0.00	0.02	[-0.05, 0.04]	44.3
BabyLM 350M	Intercept	14.55	0.01	[14.53, 14.58]	100.0
BabyLM 350M	Freq	-0.12	0.01	[-0.14, -0.09]	0.0
BabyLM 350M	Predic.	0.10	0.01	[0.07, 0.12]	100.0
BabyLM 350M	Freq:Predic.	0.00	0.01	[-0.03, 0.02]	41.6
BabyLM 1.3B	Intercept	15.30	0.03	[15.24, 15.36]	100.0
BabyLM 1.3B	Freq	-0.04	0.03	[-0.10, 0.02]	8.6
BabyLM 1.3B	Predic.	0.01	0.03	[-0.04, 0.07]	66.0
BabyLM 1.3B	Freq:Predic.	0.02	0.03	[-0.04, 0.08]	73.6

Table 6: Joint frequency × FTP model at the first layer for UP as subword. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Coef	Est.	Err.	95% CI	% > 0
OLMo-3 7B	Intercept	12.32	0.04	[12.24, 12.40]	100.0
OLMo-3 7B	Freq	-0.45	0.05	[-0.54, -0.36]	0.0
OLMo-3 7B	Predic.	-0.99	0.04	[-1.07, -0.91]	0.0
OLMo-3 7B	Freq:Predic.	-0.06	0.04	[-0.15, 0.02]	7.3
BabyLM 125M	Intercept	11.73	0.07	[11.58, 11.88]	100.0
BabyLM 125M	Freq	-0.48	0.08	[-0.62, -0.32]	0.0
BabyLM 125M	Predic.	0.16	0.08	[0.01, 0.32]	98.0
BabyLM 125M	Freq:Predic.	-0.03	0.08	[-0.18, 0.13]	36.5
BabyLM 350M	Intercept	11.23	0.07	[11.09, 11.37]	100.0
BabyLM 350M	Freq	-0.42	0.07	[-0.56, -0.28]	0.0
BabyLM 350M	Predic.	0.16	0.07	[0.02, 0.30]	98.4
BabyLM 350M	Freq:Predic.	0.06	0.08	[-0.09, 0.22]	79.4
BabyLM 1.3B	Intercept	9.91	0.06	[9.79, 10.03]	100.0
BabyLM 1.3B	Freq	-0.24	0.06	[-0.36, -0.12]	0.0
BabyLM 1.3B	Predic.	-0.23	0.06	[-0.36, -0.12]	0.0
BabyLM 1.3B	Freq:Predic.	-0.03	0.06	[-0.16, 0.09]	30.0

Table 7: Joint frequency × FTP model at the final layer for UP as subword. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

4.4 Experiment 3: Whisper

Component	Train pos	Train neg	Val pos	Val neg
Encoder	1,000	1,000	1,000	1,000
Decoder	1,000	1,000	1,000	1,000

Table 8: Classifier training and validation split sizes for Experiment 3 (Whisper). Positive examples are *word_up* instances; negatives are random non-*up* alphabetic tokens from the same segment.

Model	Coef	Est.	Err.	95% CI	% > 0
Encoder	Intercept	12.45	0.06	[12.33, 12.57]	100.0
Encoder	Freq	-0.27	0.06	[-0.39, -0.14]	0.0
Encoder	Predic.	0.31	0.06	[0.19, 0.43]	100.0
Encoder	Freq:Predic.	-0.02	0.06	[-0.14, 0.09]	33.8
Decoder	Intercept	9.24	0.02	[9.21, 9.28]	100.0
Decoder	Freq	0.04	0.02	[0.01, 0.07]	99.4
Decoder	Predic.	-0.06	0.02	[-0.09, -0.03]	0.0
Decoder	Freq:Predic.	0.00	0.02	[-0.03, 0.03]	37.7

Table 9: Joint frequency × FTP model at the first layer for Whisper encoder and decoder. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.

Model	Coef	Est.	Err.	95% CI	% > 0
Encoder	Intercept	9.19	0.03	[9.14, 9.25]	100.0
Encoder	Freq	0.11	0.03	[0.05, 0.17]	100.0
Encoder	Predic.	-0.07	0.03	[-0.13, -0.01]	1.3
Encoder	Freq:Predic.	0.03	0.03	[-0.02, 0.09]	89.6
Decoder	Intercept	8.49	0.06	[8.38, 8.59]	100.0
Decoder	Freq	-0.13	0.05	[-0.24, -0.03]	0.7
Decoder	Predic.	-0.42	0.06	[-0.53, -0.31]	0.0
Decoder	Freq:Predic.	-0.07	0.05	[-0.17, 0.03]	8.8

Table 10: Joint frequency × FTP model at the final layer for Whisper encoder and decoder. Estimates are from Bayesian linear mixed-effects models (brms); % > 0 indicates the percentage of posterior samples with a positive estimate.